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COMMENTARY

Practical guidance for validating the predictive performance in the presence of missing data: a guide for the clinical researcher

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Abstract

Background and Objectives: Prediction models are widely used across all fields of medicine as tools to support patient counseling and guide treatment decisions. A key step before any prediction model can be implemented in clinical practice is internal validation, for which principles are well described in the literature. However, the application of these principles is challenging when complex models are used or when missing values are present in the predictor variables. Approaches for internal validation and handling of missing data often result in a multitude of datasets, such as multiple bootstrapped samples across multiple imputations. Analyzing such cross-multiplied datasets in a streamlined manner is not straightforward.

Methods, Results, Conclusion: This paper provides practical guidance and a structured R workflow to support clinical researchers in combining internal validation and imputation methods when building reliable prediction models. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: Prediction; Internal validation; Imputation; Penalization; Shrinkage; Model validation; Missing data

1. Introduction

Clinical prediction models translate patient data into estimates of outcome risk to support data-driven decisions. Before a clinical prediction model can be implemented in clinical practice, its accuracy should be validated. Internal validation (ie, performance assessment on the training data itself) is the first step in model validation and strongly recommended for all model development studies [1,2]. Its goal is to simulate model robustness on out-of-sample data, thus estimating generalizability. This involves correcting for optimism, the phenomenon whereby a model appears to fit the training data more accurately than it performs in real-world applications.

Pronounced optimism is often caused by small sample sizes or overfitting, the latter entailing that the model fits idiosyncrasies of the development data rather than real patterns [3]. The evaluation of optimism by an internal validation procedure often involves resampling techniques such as bootstrapping or cross-validation, generating multiple datasets for repeated analyses [3]. When missing data are encountered, they are commonly handled using multiple imputation (under the missing [completely] at random assumption), which also generates multiple complete datasets requiring aggregation. Furthermore, some penalization techniques, aimed to reduce overfitting, also require resampling techniques to determine the shrinkage factor [4]. These challenges can make internal validation a very complex procedure, involving multiple dataset layers, which may hinder both its correct implementation and its use altogether if not approached with sufficient statistical rigor.

In statistical software, a multitude of procedures and functions are available for internal validation, multiple imputation, and penalization. Some statistical software

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What is new?

Key findings

- Prediction modeling faces three challenges: regularization, missingness, validation.
- Tackling this involves combining multiple dataset layers, complicating analyses.

What this adds to what was known?

- We provide practical guidance and a structured R workflow for clinical researchers.
- This workflow helps the clinical researcher to address the core challenges.

What is the implication and what should change now?

- Real-world data illustration shows the practical feasibility of MI-Boot.

packages even offer procedures that integrate two of these goals [5,6]. However, limited guidance is available to integrate all three goals of out-of-sample performance evaluation, imputation, and penalization.

Our principal aim is to offer practical guidance on how to combine these three procedures, focusing on the first two, which we will structure along a clinical narrative. We commence with a literature review of appropriate methods. Then, we present practical steps in a step-by-step guide along a clinical example using data from 619 patients who underwent elective esophagectomy [7]. The dataset includes demographics (age, sex), comorbidities, surgical characteristics, and tumor characteristics to predict the need for elective intensive care unit (ICU) admission following surgery. The guide is accompanied by a structured R workflow available on GitHub.

2. Methods to combine multiple imputation and bootstrapping

For small-sample prediction models, internal validation is best performed using resampling techniques such as bootstrapping, where samples are repeatedly drawn from the data with replacement, to simulate out-of-sample conditions. To handle missing data, multiple imputation is a frequently used and recommended approach which replaces missing values with plausible alternatives to generate multiple complete datasets. Analyses are then performed on each individual dataset, followed by pooling of the results to account for uncertainty attributable to the imputation of data. Both bootstrapping and multiple imputation produce multiple datasets as intermediate steps in the analysis (details in the Supplementary Materials).

To combine bootstrapping with multiple imputation, two approaches can be distinguished: multiple imputation followed by bootstrapping (MI-Boot), or bootstrapping followed by multiple imputation (Boot-MI). In the first approach, M-imputed datasets are generated, and bootstrap estimation is applied to each of them (Fig 1). In the second approach, B bootstrap samples are drawn from the original (incomplete) dataset, followed by multiple imputation of each of these samples (Boot-MI) (Fig 2).

Several studies have been conducted to investigate which approach (variations of MI-Boot or Boot-MI) yields performance measures and optimism closest to the true value. Two simulation studies found that estimates obtained by MI-Boot were optimistically biased, while estimates obtained by Boot-MI were largely unbiased to slightly pessimistically biased [8,9]. Increasing the number of bootstrap samples decreased the biases for both approaches [9]. Another study investigated how to obtain valid bootstrap inference in these scenarios [10]. It showed that both MI-Boot and Boot-MI were intuitively appealing and led to correctly estimated confidence intervals [10]. As a rule of thumb, the authors suggested that Boot-MI may be preferred for a small M or large imputation uncertainty, and MI-Boot may be preferred for moderate M and little/normal imputation uncertainty. They also show that Boot-MI is 13 times more computationally intensive.

In summary, Boot-MI appears to be the preferable choice from a theoretical standpoint. But MI-Boot has the advantage of significantly smaller computational demands, while the magnitude of bias can be controlled to a practically satisfactory level with increasing number of bootstraps and an appropriate number of imputations [9]. Note that these computational considerations become particularly important when using penalization methods where shrinkage factors must be determined. For our clinical example, we decided to use MI-Boot for its advantages in computational demands, and as we had little imputation uncertainty, we assumed that the bias would be minimal. To confirm this hypothesis, we also repeated our analysis using Boot-MI.

3. Step-by-step guide to combine multiple imputation and bootstrapping

Below, we present a practical example on how the MI-Boot framework can be applied in a real-world clinical scenario. Practical pointers and analytical scripts (R code) are available on GitHub at https://github.com/nikkidemul/MI_Boot.

To illustrate the analytical workflow, we reanalyzed data from 619 patients who underwent elective esophagectomy, of whom 380 (61%) required ICU admission postoperatively [7]. The dataset includes a comprehensive set of preoperative predictors, including demographics (age, sex), comorbidities, surgical characteristics, and tumor characteristics. The outcome of interest was the need for elective

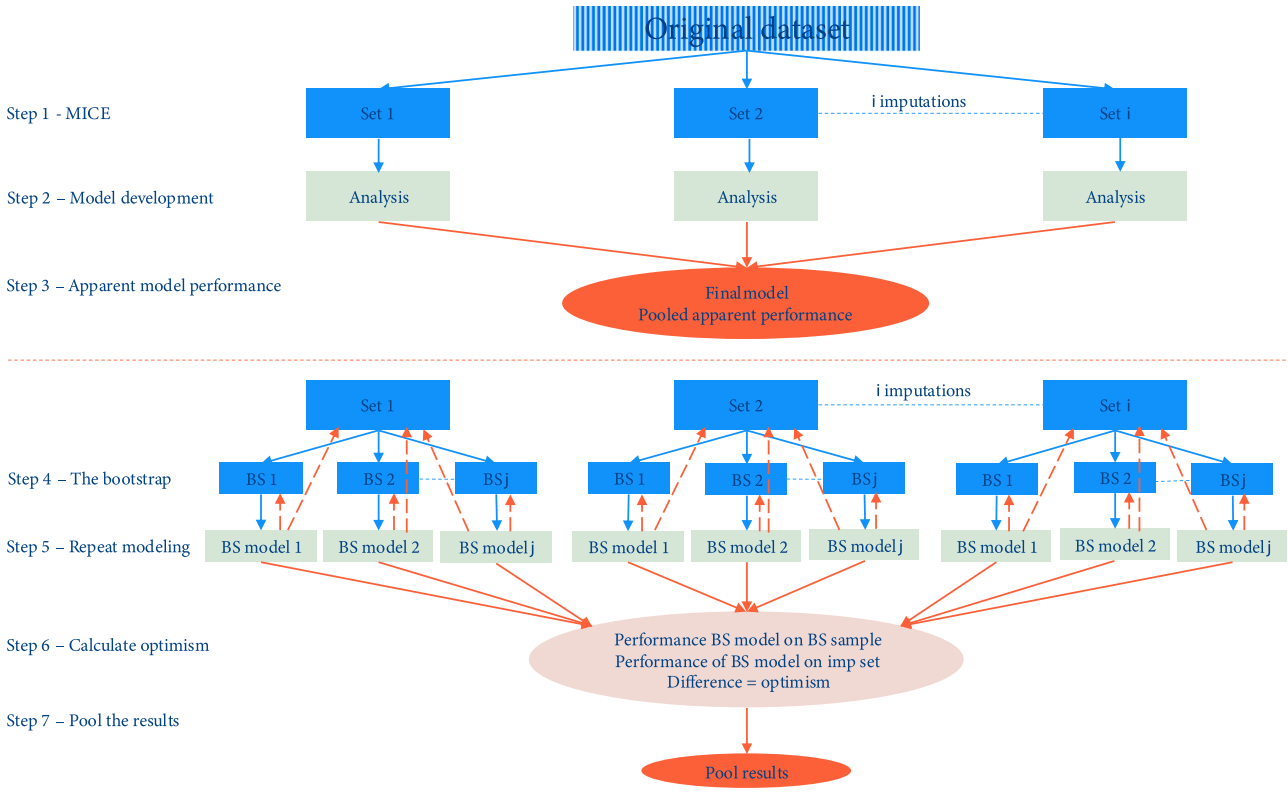

Figure 1. Schematic representation of the MI—Boot procedure. All steps are described in depth in the main text. Step 1-3 comprise of the imputation procedure, performing the analyses within each individual imputed dataset and then pooling the results. Steps 4-7 represent the goals of internal validation for optimism correction and out-of-sample performance evaluation.

ICU admission following surgery. We developed a prediction model incorporating 18 predictors (13 events per predictor). Its modest sample size, combined with a manageable level of missing data, makes this dataset particularly suitable for demonstrating our workflow.

We used the *anonymice* function from the mice R package to create a synthetic dataset with statistical properties similar to our own, enabling study replication [11]. Comparable scripts for standard (unpenalized) logistic regression are also made available on GitHub for reference. All steps are shown in Figure 1.

3.1. Step 1: multiple imputation

The first step comprises imputation of missing data [12]. In our clinical example, we used multiple imputation by chained equations; as the maximum missingness percentage for individual variables was 7%, we decided to use 10 imputed datasets and 10 iterations for the imputation procedure [2,12]. Predictive mean matching was used for continuous variables and logistic regression and polytomous logistic regression for binary and categorical variables.

3.2. Step 2: model development

Development of a prediction model across multiple imputed datasets consists of three stages:

a. Candidate predictor selection based on clinical knowledge [13].
b. Model fitting within each imputed dataset.

In this step, a model will be developed for each imputed dataset; let i, i = 1, …, I indicate the imputed datasets, yielding I models. For backward selection, this consists of a series of variable selection steps, whereas for methods that require hyperparameter tuning to determine the shrinkage factor, this involves a resampling step. For our clinical example, we chose to develop the model using Firth's penalized logistic regression analysis with backward selection, using Flic to adjust the intercept [14]. For Firth's logistic regression, resampling is not required [15,16].

c. Pooling into a final model

After step 2b, we have I models that vary in coefficient strengths and selected predictors. To derive the final pooled model across the imputed datasets, we applied a two-step method combining majority voting and the Wald test [12,17]. The majority vote preselects predictors appearing in at least 50% of the I models across I imputed datasets. The model is then refitted in each of the I imputed datasets using the majority vote-selected variables. The I models are then pooled with Rubin's rules, and the Wald statistic determines which additional variable can be dropped. The I models are refitted excluding the selected variable, and this

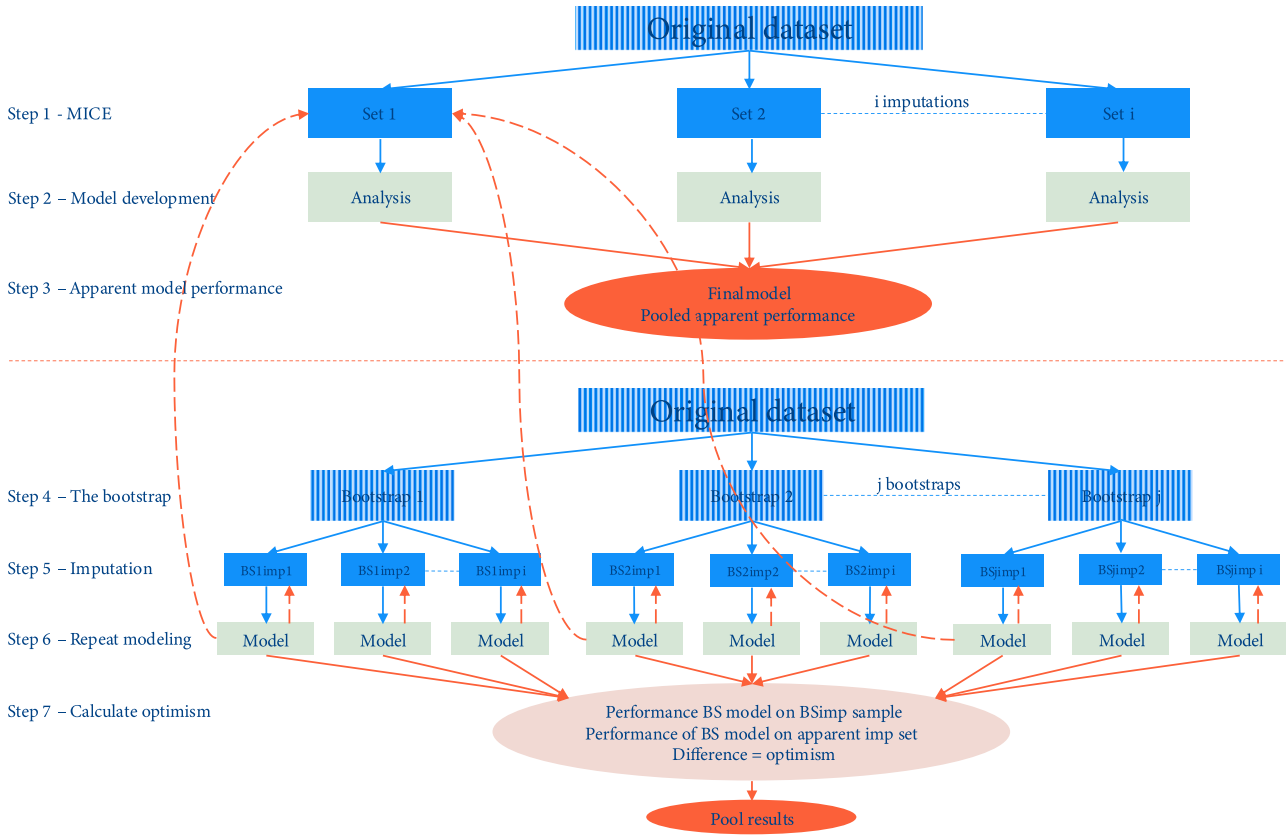

Figure 2. Schematic representation of the Boot–MI procedure. For Boot–MI, steps 1–3 are essentially the same as the MI–Boot procedure. In step 4, rather than bootstrapping the imputed datasets, bootstrap samples are drawn from the original incomplete dataset and subsequently imputed using the same imputation procedure as for step 1. The "original" bootstrap model performance is calculated the same as for MI–Boot using the sample the model was derived in. The "test" performance is calculated in the corresponding original imputed dataset. This means that the models derived in each first imputed dataset from each individual bootstrap sample are tested on the first imputation set from step 1 (as shown by the orange dashed lines), the models derived from the second imputed datasets of each bootstrap sample on the second original imputed dataset from step 1, and so forth. Optimism is defined as the difference between original and test performance (for interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

process is repeated until no further variables can be removed [12,17].

3.3. Step 3: initial model assessment

This step involves assessing the apparent performance of the final pooled model, developed in the previous step, on original data. Since the development data consist of multiple imputed datasets, the performance of the final model must be evaluated on each individual imputed dataset. This results in I performance measures, one for each imputed dataset.

From this set of performance measures, we derive a pooled point estimate of model performance using Rubin's rules. Standard errors and confidence intervals are later calculated via the bootstrap procedure.

3.4. Step 4: the bootstrap procedure

So far, the employed procedures were aimed at the optimal handling of missing data, yielding a final pooled model with apparent performance evaluated in the original data. We now turn to the goals of internal validation for optimism correction and out-of-sample performance evaluation.

For each imputed dataset, we draw J bootstrap samples (j, j = 1, …, J). This yields $I * J$ imputed, bootstrap samples. We decided on 500 bootstrap samples, resulting in 5000 datasets.

3.5. Step 5: repeat modeling

Steps 2 and 3 are repeated J times for each imputed dataset i. It is important to use the exact same steps as for apparent model development in step 2 (eg, backward selection, hyperparameter tuning) [3]. This yields $I*J$ bootstrap models. The performance of these bootstrap models is tested on the bootstrap sample they were developed in (their "original" performance) and on the imputed dataset the bootstrap sample was derived from (their "test" performance).

3.6. Step 6: calculate optimism

The optimism of performance measures is the difference between the bootstrap models' "original" performance and the bootstrap models' "test" performance. This optimism can be subtracted from the apparent model performance derived in Step 3. Of note, the optimism is estimated for each individual bootstrap model *j*, and the results pooled afterward.

3.7. Step 7: pool the results

Point estimates of performance are pooled using Rubin's rules, as in Step 3. Confidence intervals for these performance estimates are calculated as follows:

- For each imputed dataset *i*, the standard error (SE) of the point estimates across the bootstrap samples of *i* is calculated as the difference between the 0.025th and 0.975th percentiles divided by 3.92 (ie, 2 x 1.96). This yields a set of SEs, one for each imputed dataset *i*.
- These *I* SEs are pooled over the *I* imputed datasets using Rubin's rules.
- We use these final pooled SEs to compute 95% confidence intervals for apparent and corrected model performance estimates.

3.8. Step 8: report final model parameters and performance

At the conclusion of the stepwise procedure, we have obtained the final model along with a table containing apparent and corrected performance measures and their confidence intervals (Table 1). Our model had moderate performance (c-statistic 0.66 [95% confidence interval (CI) 0.61−0.70], rescaled Brier score 0.11 [95% CI 0.02−0.12], and good calibration). Optimism was substantial, leading to a significant correction in calibration measures (corrected calibration slope 0.76 [95% CI 0.72−0.80]).

4. Bootstrapping followed by MI

When using a Boot-MI approach, the same step-by-step guide as described above can be used, with some alterations. Steps 1-3 and 6−8 are the same. The bootstrap procedure (step 4) is performed on the original data, resulting in *J* bootstrap samples still containing missing data. Multiple imputation is then applied to each of these *J* bootstrap samples, resulting in *J*I* complete datasets in which *J*I* bootstrap models can be developed (step 5). The "original" performance of the bootstrap models is tested on the sample these models were developed in. The "test" performance is estimated on the respective imputed sample from step 1 (eg, model j_1 is tested on imputation set 1, model j_2 on imputation set 2, etc., Fig 2). For our clinical example, the results obtained using a Boot-MI approach were similar (Supplementary Materials), with computational time required for Boot-MI roughly 15 times longer than that for MI-Boot.

5. Methodological considerations

Prediction modeling guidelines emphasize the importance of appropriate handling of missing data and penalization and internal validation to reduce and assess model optimism. We provide a step-by-step guide for internal validation in the presence of missing data, applying penalization techniques and enabling clinical researchers to conduct their analyses efficiently. Combining these methods is therefore beneficial, as it avoids suboptimal alternatives such as performing a complete-case analysis—which, in our example case, would have excluded 20% of an already limited sample size—or foregoing internal validation altogether. Our step-by-step guide remains applicable across different modeling, pooling, or internal validation strategies.

Penalization is often applied to reduce overfitting in the model development stage, but it should also be embedded in the internal validation step. In our clinical example, we

Table 1. Performance of the final model

Performance measure	Apparent (95% CI)	Original (BS)	Test (BS)	Optimism	Corrected (95% CI)
C-statistic	0.66 (0.61−0.70)	0.69	0.64	0.04	0.62 (0.57−0.66)
O:E ratio	1.00 (1.00−1.00)	1.00	1.00	0.00	1.00 (1.00−1.00)
Calibration intercept	−0.01 (−0.03 to 0.02)	−0.01	0.09	−0.09	0.09 (0.07−0.11)
Calibration slope	1.01 (0.98−1.05)	1.02	0.76	0.25	0.76 (0.72−0.80)
Brier	0.22 (0.21−0.24)	0.21	0.23	−0.01	0.23 (0.22−0.25)
Rescaled Brier	0.07 (0.02−0.12)	0.11	0.06	0.05	0.02 (−0.02 to 0.07)
IPA	0.07 (0.02−0.12)	0.11	0.06	0.05	0.02 (−0.02 to 0.07)

IPA, Index of Prediction Accuracy.

Results of the bootstrapping procedure for the final model (500 iterations). Apparent corresponds to the apparent predictive performance measures (uncorrected) on development data. Original indicates the performance of the bootstrap models applied to the bootstrap samples, test indicates the performance obtained from the bootstrap models applied to their corresponding original imputed datasets. Corrected indicates the values after optimism correction. The rescaled Brier score, or Brier skill score, rescales the Brier score to a range of 0 (worst) and 1 (best), adjusting for the average risk for each patient (ie, the "null model" without any predictors).

chose Firth's logistic regression with backward selection, a preferred method for small datasets and datasets that deal with separation, which also stabilizes the bootstrap procedure [15]. Other modeling strategies can be incorporated in step 2b of our guide. Penalization techniques such as lasso and ridge regression all require cross-validation to determine the hyperparameter [18]. As the entire procedure for model development (including tuning of the hyperparameter) should also be repeated in step 5, this is obviously computationally intensive.

In our clinical example, multiple imputation, variable selection, and internal validation had to be addressed jointly, illustrating how our guide can deal with this in practice. For the selection of predictors for the initial model fit (step 2), we chose a two-step approach: a majority vote, followed by the Wald test. Three main approaches exist for variable selection in multiply imputed data: (1) majority voting, (2) the stacked method (ie, combining all imputed datasets into one, assigning a fixed weight to each record i and applying standard selection methods), and (3) the Wald test, used in Prediction Model Pooling, Selection and Performance Evaluation Across Multiply Imputed Datasets [12]. For the latter, the Wald test iteratively identifies which variable to remove from the pooled model, whereafter the models are refitted on each imputed dataset I, excluding the predictor that should be dropped. Although this method is preferred, it is computationally intensive. Therefore, a two-step approach was proposed combining majority voting for preselection, with the Wald test to enhance model efficiency [17]. We compared the Wald test to the two-step method and found identical models in terms of selected variables and SEs (data not shown). Thus, we chose the more efficient two-step approach, though any of the mentioned methods can be used in step 2c of the step-by-step guide.

Alternative resampling methods for internal validation include cross-validation and the 0.632(+) bootstrap (see Supplementary Terminology box for details) [8]. Steps 1−3 of the step-by-step guide apply to these methods without modification. For the MI-Boot approach, resampling occurs in step 4, with optimism calculated as the difference between model performance in the validation and development sets [8]. Note that there are several approaches to also correct the confidence intervals based on resampling, which may yield more accurate coverage [19]. For cross-validation, a Boot-MI approach is recommended, which can also be applied to the 0.632(+) bootstrap [20]. A schematic representation of this workflow is included in the Supplementary Files.

In conclusion, all prediction models should be validated, most real-world datasets are small enough to require penalization, and many contain missing predictor values. As a result, developing well-validated models typically involves multiple dataset layers, requiring a streamlined analysis approach. MI-Boot is computationally more efficient and maintains low bias with sufficient bootstraps. Our step-by-step guide supports clinical researchers in navigating these challenges by integrating imputation, penalization, and internal validation into a coherent workflow, applicable to diverse modeling contexts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

N. De Mul: Writing − review & editing, Writing − original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **V.M.T. de Jong:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Conceptualization. **O.L. Cremer:** Writing − review & editing, Supervision, Methodology, Conceptualization. **H.J.S. de Grooth:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Conceptualization. **S. el Bouhaddani:** Writing − review & editing, Writing − original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112159.

Data availability

The script is available from GitHub (https://github.com/nikkidemul/MI_Boot). A synthetic dataset supporting the findings of this study is freely available from the corresponding author upon reasonable request.

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